Coffee Shop Staffing Optimizer

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Why Coffee Shop Staffing Optimizing Matters

- Efficient staffing is one of the hardest daily problems for small service businesses like cafés, restaurants, and bakeries.
- Demand fluctuates hour by hour; mornings are busy, afternoons slow and yet managers often schedule based on intuition rather than data.
- A **Coffee Shop Staffing Optimizer** uses mixed integer linear programming to generate cost effective, demand aligned schedules automatically. This project matters because it helps small businesses reduce costs and get more efficient.

Benefits of Optimizing Scheduling for Cafes

- Reduce Labor Costs: Labor is typically 30–40% of total expenses. Optimizing shifts prevents overstaffing during slow hours while ensuring enough coverage during rush times.
- Improve Service Quality: By aligning staff availability with peak demand, wait times and order errors drop, improving customer satisfaction and repeat sales.
- Save Manager Time: Owners or managers often spend hours manually building schedules; an optimization model can produce a fair, compliant schedule in seconds.

Parameters

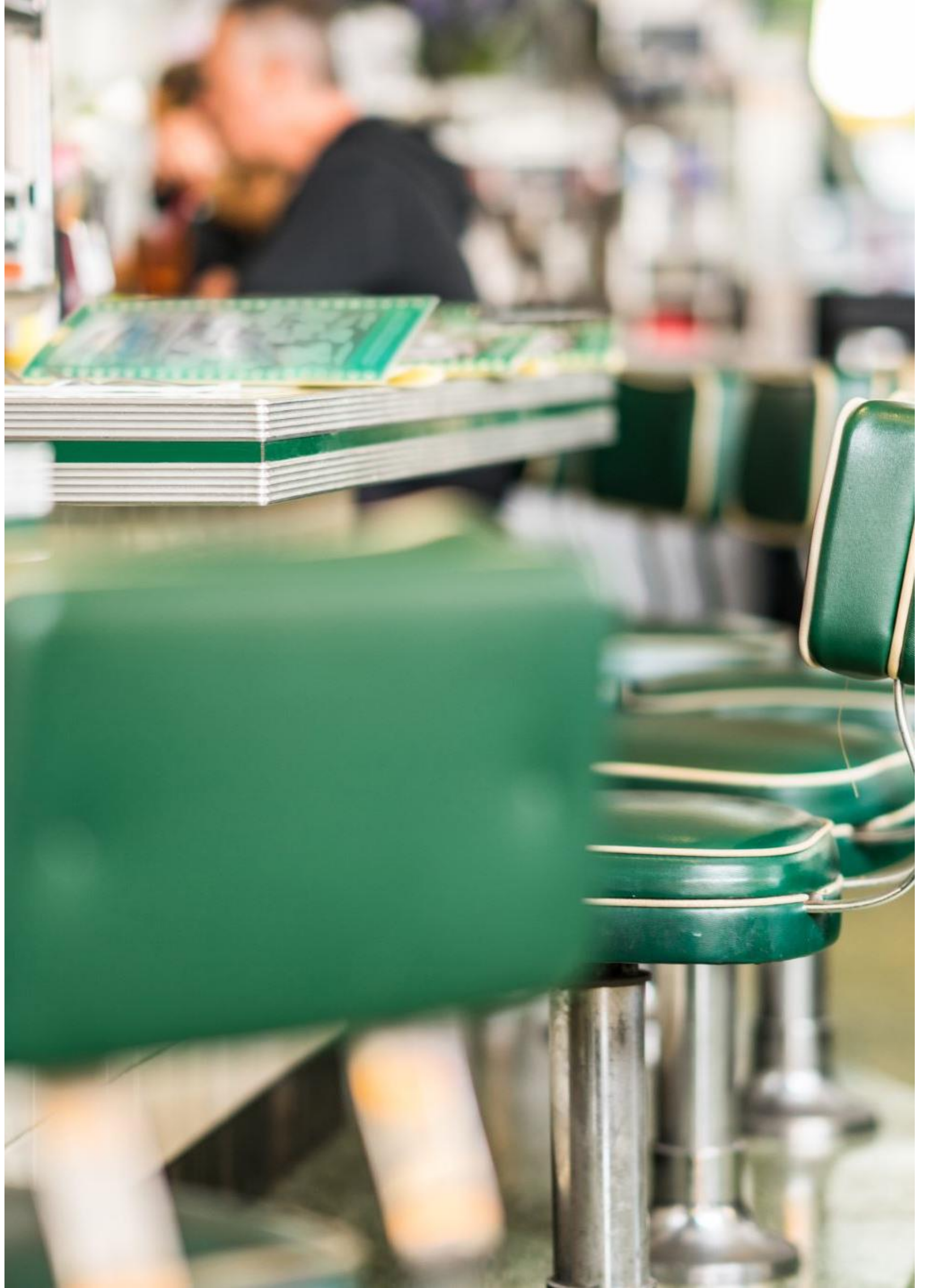
- E: set of employees
- H: set of working hours (e.g., 8 AM – 9 PM)
- w_e :hourly wage of employee e
- d_h :required number of baristas at hour h
- $A_{e,h}$: 1 if employee e is available at hour h, else 0
- M_e :maximum hours employee e can work per day

Constraints for the Optimizing problem

- Meet customer demand: Each hour must have enough staff working to serve customers. Weekend and weekday schedule and demand differs so we should account for that

- Limit total hours per person: Each barista should have a cap on how many hours they can work in a day and work in a week.

- Respect employee Availability: Employees are not scheduled during unavailable hours

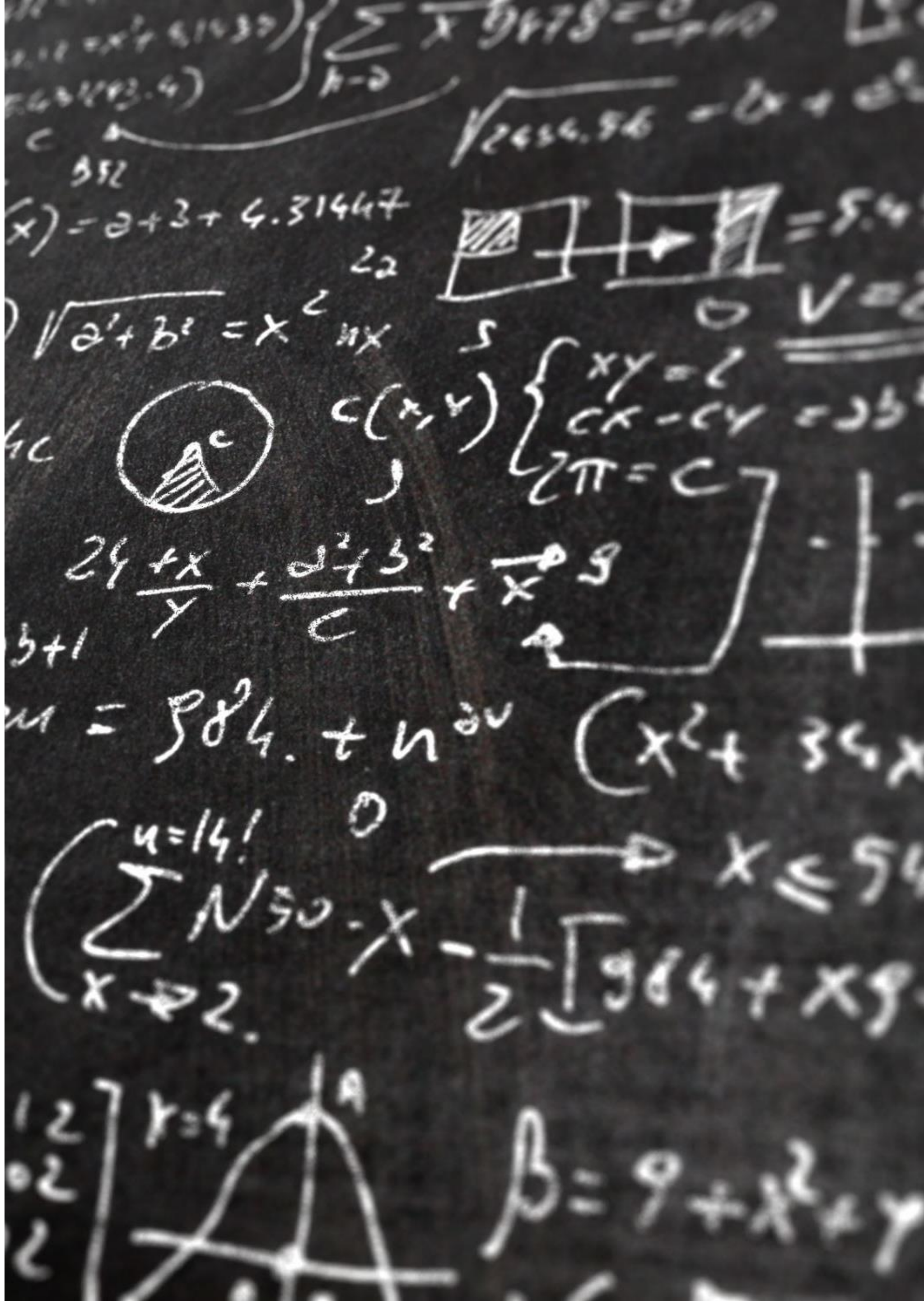


Decision Variables and Objective Function

The Decision Variable for this staffing problem is if an employee is staffed to work at a specific hour of the day.

The Objective function would be to minimize costs and minimize unmet demand after satisfying all the constraints explained in the previous slides.

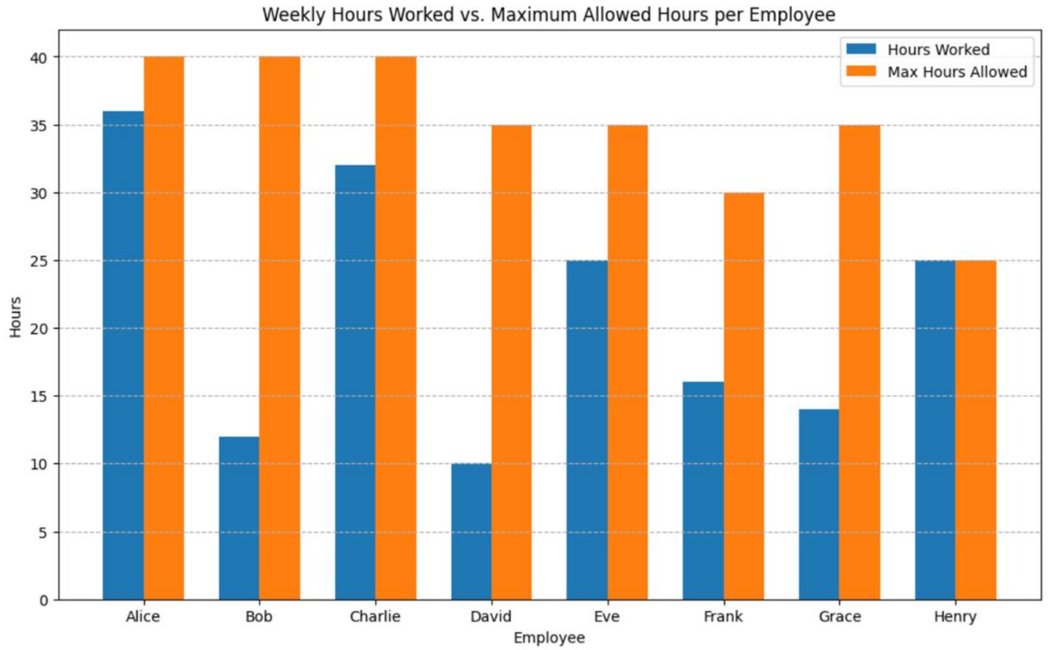
For demonstration purposes, we can assume we have 8 employees and each have certain hours of availability and maximum hours they can work which vary. We also assume the penalty for unmet demand is $500 per hours.

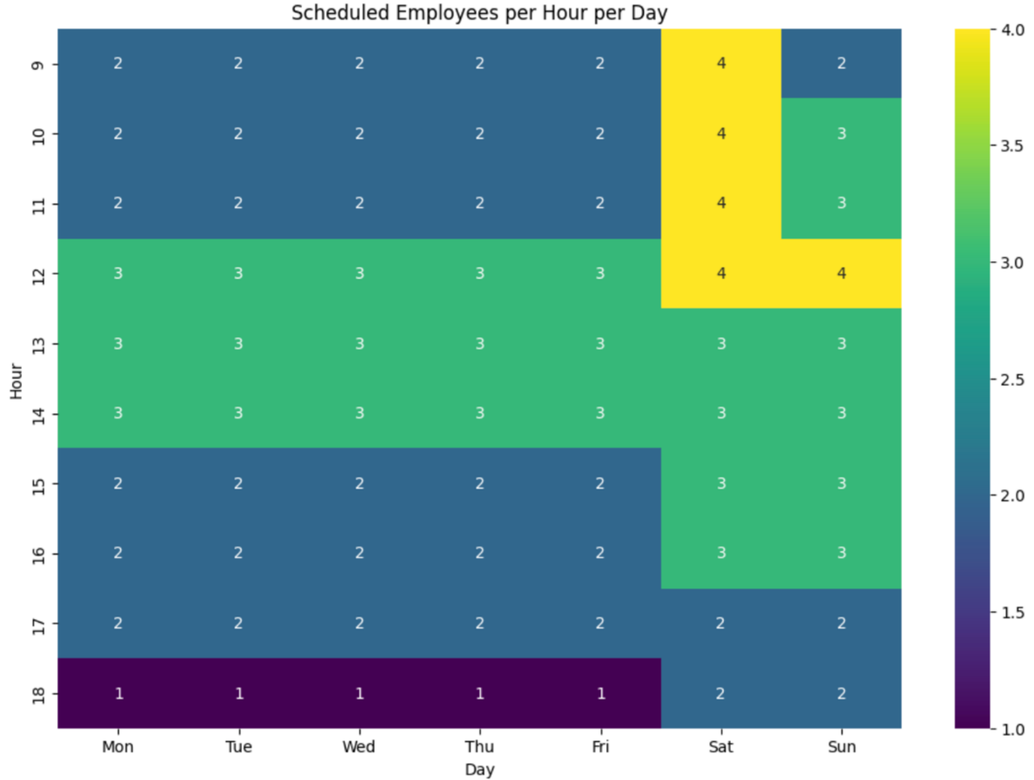


Results

We have a dataset of 5 employees; Alice, Bob, Charlie, David, Eve, Frank, Grace and Henry. In this instance, the cost for scheduling 8 individuals was $5,443.00 including the penalty of having 4 hours not sufficiently staffed.

Link to python code:
https://colab.research.google.com/drive/1qjOrCTkWxqpA9ryvrEeg3LGwc4vRZYL8?usp=sharing





Limitations

- There was a penalty set for each hour that did not have enough coverage, which was lost revenue. Due to having strict constraints, in the example we had to be understaffed for certain hours of the day which in future could be relaxed so we don't have unmet demand
- Additionally, we can carry out further sensitivity analysis to understand the tradeoff between unmet demand and scheduling costs. For instance, if we increase penalty and decrease unmet demand to zero, we would need to understand how customer satisfaction could offset the increase in labor cost.

THANK YOU!